

# Seth Emond

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## **SUMMARY**

Mechanical Engineer with prototyping expertise and SolidWorks certification seeking design-focused role. Will relocate at own expense.

## **SKILLS**

CAD: SolidWorks (CSWA Certification), Pro/E Creo, Alibre Atom3D, Fusion360

Manufacturing & Prototyping: 3D printing, CNC routing, FEA, DFM, DFA

Technical: Python, MATLAB, Arduino, project management, technical documentation

## **PROJECTS**

### **Volatile Organic Compounds (VOC) Filtration System for 3D Printer**

- Addressed health risks from 3D printing VOC emissions in home workshop by researching particle sizes and chemical composition, designing filtration system using off-the-shelf fan/filter with custom brackets
- Designed two-part mounting bracket assembly in Alibre Atom3D featuring over-center clamping mechanism and bayonet twist-lock interface; validated mechanical function through 3D printed FDM prototype

### **Custom Filament Spool Holder with Weight Sensor**

- Eliminated print failures from filament runout by designing real-time weight monitoring system with lower friction than existing solutions
- Designed mechanical assembly integrating press-fit bearings, 3D-printed components, and load cell instrumentation; performed calibration testing and validated measurement accuracy under load conditions

### **Modular Tool-Free Drawer Organizer System**

- Designed tool-free modular organizer system featuring 3D-printed elastic compression joints and CNC-cut panels, enabling rapid reconfiguration without fasteners
- Developed Python-based parametric FEA optimization script automating simulation loops; validated design through physical testing, identified interface failure mode, and redesigned with simplified press-fit geometry

## **EXPERIENCE**

### **Project Engineer | SonTech | Rapid City, SD | 2023 – Present (internal title: Network Administrator)**

- Achieved 90%+ on-budget performance across 20+ concurrent projects through accurate design-phase cost estimation, component selection optimization, and BOM development
- Designed 50+ commercial systems including component placement, mounting solutions, and physical integration; created detailed mechanical layouts and assembly specifications
- Created engineering documentation packages in Bluebeam including system drawings, component specifications, mounting details, and installation procedures, reducing field rework by 80-90%

### **Team Lead | SonTech | Rapid City, SD | 2020 – 2023**

- Executed design adaptations for as-built field conditions (structural clearances, mounting constraints, access limitations) while preserving performance specifications
- Developed comprehensive documentation system (as-built drawings, topology diagrams, installation notes), reducing system troubleshooting time by up to 80% through organized technical records

## **EDUCATION**

Purdue University | West Lafayette, IN | 2020

Bachelor of Science in Mechanical Engineering | FE Exam